Placing Defects Where You Ask: Generating Defects at Chosen Sites in a Model Glass with a Diffusion Model

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Abstract

Defects control many properties of glasses. To study a particular defect, researchers today either wait for it to appear by chance in a slow melt-and-cool simulation, or insert it by hand, which can leave unnatural surroundings. Diffusion models can generate realistic glass structures, but cannot be told to put a given defect at a given place. We build such a model for a two-dimensional model glass. A graph neural network that respects the symmetries of the glass learns to remove noise from atom positions; told which atoms to keep fixed and which defect to create, it rebuilds a small region around a chosen site inside an existing glass. We relax every result before checking it and compare each new defect's surroundings with those of natural defects. Over three training runs, the model creates atoms with too few neighbours 3.5 times as often as hand insertion (0.22 vs. 0.06), with near-natural surroundings. It creates B-rich defects 1.5–2.4 times as often (0.45–0.71 vs. 0.30), with surroundings that are as natural or more natural. Melting and cooling the same region again almost never works. No method can create atoms with too many neighbours, and we show why. These results hold under every change to the evaluation that we tried. We also find that the number of sampling steps acts like a cooling rate, that the model must be trained on all noise levels used in sampling, and that training runs with the same validation loss follow the defect label with different strength.

1 Introduction

Glasses have no regular crystal lattice, but they still contain defects: small groups of atoms arranged differently from the rest. How a defect behaves depends on the atoms around it, so studies of how glasses deform, age or absorb light need many realistic examples of the same kind of defect. There are two common ways to get them. In a melt-and-cool (melt-quench) molecular dynamics (MD) simulation, one waits for the defect to form by chance, at a place one cannot choose. In hand insertion, atoms are moved or replaced by hand and the structure is relaxed, which can leave unnatural surroundings.

Generative models offer a third way. DM2 [1] learns to turn random atom positions into amorphous silica and metallic glasses that match MD structures, at a fraction of the cost of slow cooling. Like most generative models for materials [2–6], it is controlled by properties of the whole sample, such as the cooling rate. It cannot be asked for a particular defect at a particular place.

We treat this problem like image inpainting [7], where a model fills in a missing part of a picture. Here the model fills in a small region of a glass and is told which defect to put at its centre. Two choices guide the study. First, the region is cut out of an existing glass and everything outside it stays fixed (Task B). Asking for a defect at a location in an empty box would be trivial, because any natural defect could simply be moved there. Second, a generated defect only counts if it is still there after the whole structure has been relaxed to the nearest energy minimum, and its surroundings are compared with those of defects that formed naturally.

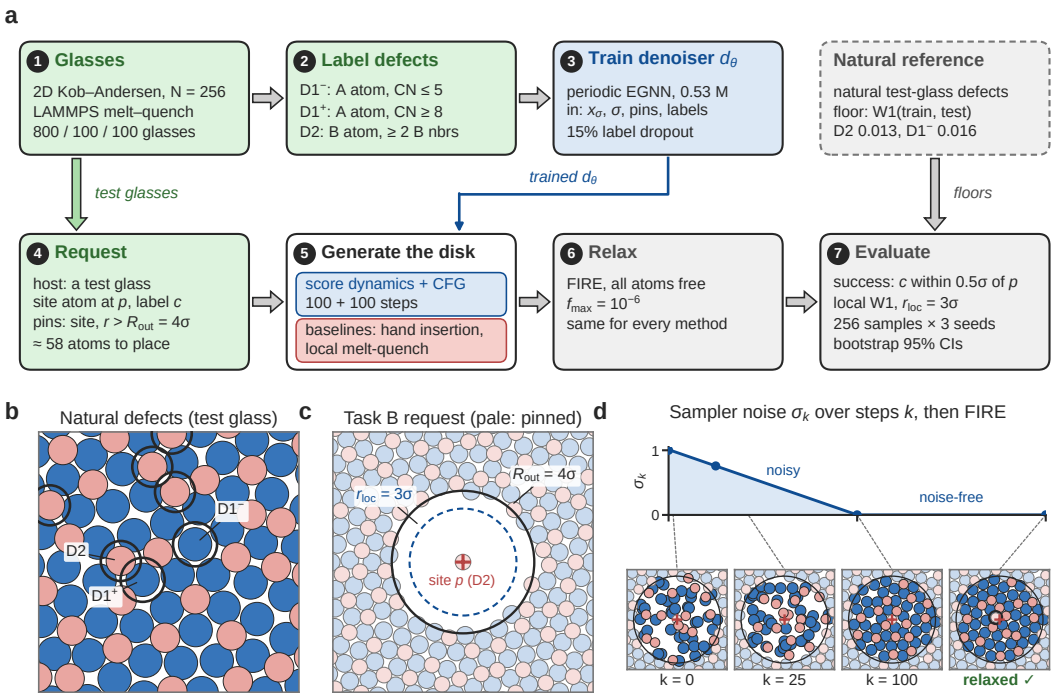


Figure 1: Overview of the study. **(a)** Workflow. We make glasses by melting and cooling (1), label their defects (2) and train the network (3). A request (4) names a glass, a site and a defect type. The model or a baseline fills in the region around the site (5). Every result is relaxed with all atoms free to move (6) and then scored (7). **(b)** Natural defects (rings) in a test glass. Large blue atoms are type A; small pink atoms are type B. **(c)** A real request. Pale atoms are held fixed, including the atom at the site (under the cross). Atoms inside the solid circle (R_{out}) are rebuilt; the dashed circle (r_{loc}) is the area used to judge realism. **(d)** The model filling this region ($w = 6$). Atoms start at random positions and are cleaned up as the noise level σ_k falls. After relaxation, the requested D2 defect (ring) sits at the site.

Contributions.

1. A test for placing defects in glasses: fill a region inside an existing glass, check the defect after relaxation, compare its surroundings with natural defects, and compare with an MD baseline that melts and cools the same region again.
2. A diffusion model built on a periodic equivariant graph neural network (EGNN). It takes atoms held fixed, a defect label for each atom and classifier-free guidance [8]. It beats hand insertion and local melt-quench for two of three defect types, across evaluation settings, repeated sampling and three training runs.
3. Practical lessons: sampling steps act as a cooling rate; training must cover all noise levels used in sampling; runs with the same loss follow the label with different strength; built-in symmetry matters more than a low loss; and atoms with too many neighbours depend on atoms more than 3.5σ away.

Figure 1 gives an overview.

2 Related work

Diffusion models for materials. Diffusion models [9, 10] learn to turn noise into data step by step. They now generate crystals [2–4] and disordered solids. For glasses, DM2 [1] learns to remove random shifts from atom positions and builds new glasses by repeating this clean-up with small random kicks (score dynamics). It can be told the cooling rate and reproduces the structure and mechanical properties of silica and metallic glasses. Other work steers generation with measured spectra [6] or target properties [5]. All of these control properties of the whole sample. None of them

places a chosen local structure at a chosen site.

Local control. In images, RePaint [7] fills a masked region using a model trained without labels, and guidance [8, 11] makes a model follow a label more strongly while sampling. In materials and molecules, the closest work fills in parts of crystals around a fixed host [12], relaxes crystals that contain given point defects [13], and links fixed molecule fragments [14]. Glasses differ in two ways. There is no lattice to compare a defect with, so a defect is a local arrangement that also appears naturally. And realism can only be judged by comparing with many natural examples. Our evaluation is built on both points.

3 Setup

Glass and defects. We use the two-dimensional Kob–Andersen glass [15, 16], a standard model with two kinds of atoms, A (large, 65%) and B (small, 35%), that interact through Lennard-Jones forces. Each glass has $N = 256$ atoms in a square box of side 14.6σ that repeats in both directions (periodic boundaries); σ, the size of an A atom, is our unit of length. We make glasses in LAMMPS [17]: melt at temperature $T = 2.0$ for 100 time units, cool by 10^{-2} per time unit to $T = 0.01$, and minimise the energy. We keep 800 training, 100 validation and 100 test glasses, and discard any that crystallise. An atom's neighbours are the atoms within 1.4 times the contact distance of the two atom types. We define three defect types:

- **D1**$^-$: an A atom with at most 5 neighbours (1.7 per glass on average);
- **D1**$^+$: an A atom with at least 8 neighbours (1.4 per glass);
- **D2**: a B atom with at least two B neighbours within 1.2σ (10 per glass).

These rules were set using the training glasses and never changed afterwards.

Tasks. In **Task B**, our main task, we take a test glass and choose a site: the atom of the right type nearest a random point. That atom, which carries the defect label, and every atom more than $R_{\text{out}} = 4\sigma$ from it are held fixed. The ≈58 atoms in between are removed and rebuilt from random starting positions. In **Task A**, the model builds a whole new box with the defect at its centre. We report it for completeness, since choosing a location means little in an empty box.

Model architecture. The network $d_\theta(x_\sigma, \sigma, m, c)$ (Figure 2a) takes noisy atom positions x_σ and returns one 2D arrow per atom: its guess of how far the noise has pushed that atom. If the glass is shifted or rotated, or its atoms are renumbered, these arrows should shift, rotate or be renumbered in the same way. Instead of hoping the network learns this, we build it in, using an E(n)-equivariant graph neural network (EGNN) [18] that we adapted to periodic boxes. It has four parts.

- **Graph.** Each atom is a node, linked to every atom within $r_c = 2.5\sigma$ (about 24 neighbours). Distances are measured across the box edge when that is shorter, so atoms near opposite edges can be neighbours. Each distance is described by 16 smooth bump functions (radial basis functions) and multiplied by a cut-off f_c that fades to zero at r_c, so links switch on smoothly as atoms come close.
- **Atom features.** Each atom starts with a vector of 128 numbers, h_i^0, built from its type t_i, whether it is held fixed (m_i), its label c_i and the noise level σ. The label is "none" everywhere except at the site, where it names the requested defect. The noise level tells the network how far atoms may be from their clean positions.
- **EGNN layers.** Four layers (Figure 2b) each pass messages between neighbours, update the features and produce an arrow for each atom:

$$
\begin{aligned}
m_{ij} &= \phi_e\big(h_i, h_j, \mathrm{rbf}(|r_{ij}|)\big)\, f_c(|r_{ij}|), \\
h_i &\leftarrow h_i + \phi_h\Big(\mathrm{LN}(h_i),\ \tfrac{1}{K_0}\textstyle\sum_j g(m_{ij})\, m_{ij}\Big), \\
v_i^\ell &= \tfrac{1}{K_0}\textstyle\sum_j r_{ij}\, \phi_x(m_{ij})\, f_c(|r_{ij}|).
\end{aligned}
\tag{1}
$$

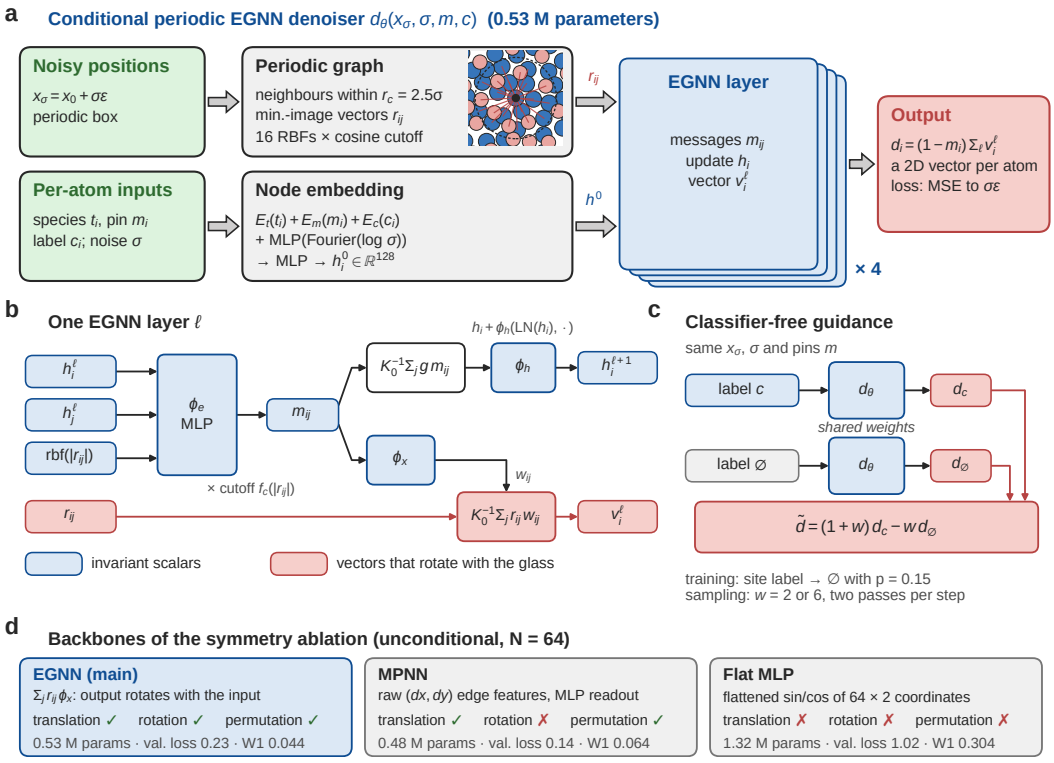


Figure 2: The neural networks. **(a)** The denoising network. Noisy positions define a graph of neighbouring atoms, the other inputs define each atom's starting features, and four EGNN layers turn both into one arrow per atom. **(b)** One EGNN layer (Eq. 1). Blue items do not change when the glass is rotated; red items are vectors that rotate with it. Each output arrow is a sum of neighbour vectors with rotation-free weights, so the prediction rotates exactly with the glass. **(c)** Classifier-free guidance runs the same network with and without the site label and combines the two answers. **(d)** The three networks compared in the symmetry test (Table 3).

Here r_{ij} is the vector from atom i to its neighbour j; ϕ_e, ϕ_h and ϕ_x are small two-layer neural networks; g is a gate between 0 and 1; LN is layer normalisation; and $K_0 = 24$ is the typical number of neighbours. Messages and features depend only on distances, so they do not change when the glass is rotated. The only directions come from the vectors r_{ij}, so each output arrow is a weighted sum of them and turns exactly with the glass. Unlike the original EGNN, we do not move atoms inside the network; each layer simply adds its arrow to the output.

- **Output.** The prediction for atom i is $d_i = (1 - m_i)\sum_\ell v_i^\ell$, which is zero for held atoms. Our tests (`tests/test_equivariance.py`) check numerically that shifts, rotations and renumbering act on the output as they should.

The network has 0.53 million parameters, 59 thousand of them in the atom features. Because the request enters only through atom features, the same network handles every kind of request. Guidance (Figure 2c) needs no second network: we run the same network with and without the site label.

Networks for the symmetry test. To see how much the built-in symmetry helps (Section 4.4), we also trained two simpler networks with the same inputs and settings (Figure 2d). The message-passing network (MPNN) [19] uses the same graph and messages as the EGNN, but also receives the raw x and y parts of each neighbour vector and predicts the arrows directly from the atom features with a small network. It therefore handles renumbering correctly but not rotation (0.48 million parameters). The flat MLP has no graph. For all 64 atoms, sorted by type, it lists the sine and cosine of each coordinate across the box, the atom type, the fixed flag and the label, adds the noise level, and passes everything through four layers of 512 units to 128 outputs (1.32 million parameters). It respects the repeating box and nothing else.

Training. We follow DM2 [1]. We add random noise to the free atoms, $x_\sigma = x_0 + \sigma\varepsilon$ with ε drawn from a standard normal distribution, and train the network to predict the added shift $\sigma\varepsilon$. The loss is the mean squared error over the free atoms U:

$$\mathscr{L}(\theta) = \mathbb{E}_{x_0,\sigma,\varepsilon,m,c}\left[\frac{1}{|U|}\sum_{i\in U}\left\|d_\theta(x_\sigma,\sigma,m,c)_i - \sigma\varepsilon_i\right\|^2\right], \tag{2}$$

where m marks the held atoms and c holds the labels. Other details:

- the noise level σ is drawn uniformly between 10^{-3} and $\sigma_{\max}$;
- training examples are whole boxes (50%), small patches (30%) or filled-in regions (20%), centred on a natural defect half of the time;
- the label is hidden 15% of the time, which classifier-free guidance needs;
- AdamW optimiser, learning rate 2×10^{-4}, batch size 16, gradient clipping at 1, 50,000 steps, and a moving average of the weights (0.999); training takes about 4 hours on one T4 GPU.

We use two settings that differ only in $\sigma_{\max}$: 0.5 (DM2's choice, half a bond length; used for D1$^-$ and Task A) and 1.0 (Section 4.2; used for D2).

Sampling. DM2's score dynamics starts from random positions and repeats two moves: subtract the predicted shift, then add a smaller random kick,

$$x_{k+1} = x_k - \tilde{d}(x_k,\sigma_k) - \eta_k, \qquad \eta_k \sim \mathscr{N}(0, s_k^2 I), \quad s_k \sim \mathscr{U}(10^{-3},\sigma_k). \tag{3}$$

Held atoms are put back after every step. Classifier-free guidance [8] combines the predictions with and without the label,

$$\tilde{d} = (1+w)\,d_\theta(x,\sigma,m,c) - w\,d_\theta(x,\sigma,m,\varnothing). \tag{4}$$

The guidance weight w sets how strongly the model follows the label; $w=0$ adds no extra push. We run 100 steps with σ_k falling evenly from 1.0 to 10^{-3}, then 100 steps without noise. The noise level given to the network is capped at the $\sigma_{\max}$ it was trained on.

Baselines.

- *Chance*: fill the same region with no label.
- *Hand insertion* (common practice): take a natural defect and its neighbours within 1.5σ from a training glass and paste it at the site; each pasted atom replaces the nearest host atom of the same type.
- *Local melt-quench*: in LAMMPS, freeze the atoms beyond R_{out}, then melt and cool the region at the rate used to make the dataset. This is waiting for the defect, but only inside the region.
- For Task A: *Clamp* and *RePaint* [7], which use a model trained without labels.

Evaluation.

- **Relaxation.** Every result is relaxed with the FIRE algorithm [20] until the largest force is below 10^{-6}. All atoms can move, including the host.
- **Success.** The atom of the requested type nearest the site must lie within $r_{\text{tol}} = 0.5\sigma$ of it and must be a defect of the requested type.
- **Local realism (local W1).** We compare the energies of the atoms within $r_{\text{loc}} = 3\sigma$ of the site with the same energies around natural defects of that type in the test glasses, using the Wasserstein-1 distance (W1). Lower is more natural. Natural defects from the training glasses, compared with those from the test glasses, score D1$^-$ 0.016, D1$^+$ 0.018 and D2 0.013. We call this the natural floor: the best a method can expect.
- **Statistics.** 256 samples per result, with 95% confidence intervals (CIs) from bootstrap resampling.

4 Results

4.1 Main result: defects inside existing glasses

Table 1 and Figure 4 give the Task B results. Model results are averages ± standard deviations over three training runs (seeds 0–2), each with 256 samples.

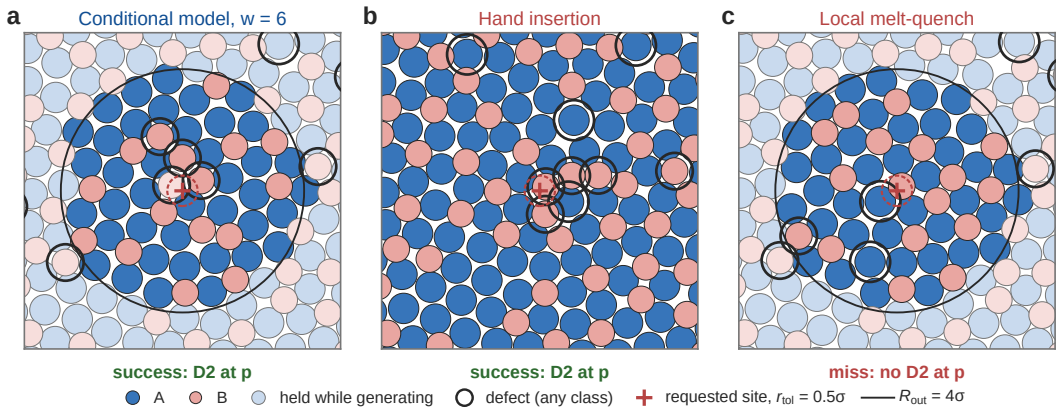


Figure 3: The request of Figure 1c answered by each method, after relaxation. The view is $\pm 5\sigma$ around the site; pale atoms were held fixed during generation; rings mark defects. **(a)** The model places a D2 defect at the site. **(b)** Hand insertion also works here, but on average it leaves less natural surroundings. **(c)** Local melt-quench rebuilds the region without a D2 at the site. Across the six requests made for these figures, the model succeeded twice, hand insertion four times and local melt-quench once; Table 1 gives the rates over 256 requests per training run.

Table 1: Task B: success after relaxation and local realism (W1 of atom energies within 3σ of the site; lower is more natural). Model rows: mean $\pm$ standard deviation over three training runs. Baselines: estimate with 95% CI. D2 uses $\sigma_{max} = 1.0$; D1$^-$ and D1$^+$ use $\sigma_{max} = 0.5$. †2,900-step sampler. ‡Seed 0 only.

	D2 (B-rich)		D1$^-$ (too few neighbours)		D1$^+$
Method	success	local W1	success	local W1	success
Chance (no label)†	0.03	0.062	0.02	0.053	0.00
Local melt-quench	0.05 [0.03, 0.08]	0.065	0.00 [0.00, 0.00]	0.024	0.01
Hand insertion	0.30 [0.24, 0.36]	0.087	0.06 [0.04, 0.09]	0.032	0.02
Model, $w = 2$	0.45 $\pm$ 0.11	**0.051** $\pm$ 0.006	**0.22** $\pm$ 0.02	**0.022** $\pm$ 0.002	0.03‡
Model, $w = 6$	**0.71** $\pm$ 0.08	0.072 $\pm$ 0.016	0.21 $\pm$ 0.04	0.038 $\pm$ 0.004	0.04‡
Natural floor	—	0.013	—	0.016	—

The model beats both baselines for the two defect types that can be placed.

- **D2.** At $w = 6$ the model succeeds 2.4 times as often as hand insertion on average (2.1–2.7 times for each training run), with surroundings as natural or more natural. At $w = 2$ it succeeds 1.5 times as often (1.2–1.9 times), with clearly more natural surroundings for every run (W1 0.045–0.056 vs. 0.087).
- **D1$^-$.** The model succeeds 3.5 times as often as hand insertion at $w = 2$, and its surroundings score 1.2–1.5 times the natural floor.
- **Local melt-quench** almost never creates the defect at the site. Melting and cooling a 60-atom region at the dataset's cooling rate is simply waiting for the defect on a small scale, and rare defects stay rare.
- **Cost.** Each success takes the model 1–4 GPU-seconds. This is similar to hand insertion (0.8–3.6 s, nearly all of it relaxation) and much less than local melt-quench (65 s per D2 success, and no D1$^-$ success at all).

Stronger guidance gives more successes but less natural surroundings. With stronger guidance, the defect is in place before relaxation more often: for D2, in 0.86 of samples at $w = 2$ and 1.00 at $w = 6$ (seed 0). More of these defects also survive relaxation, but their surroundings become less natural. For D1$^-$, success stops improving after $w = 2$: extra guidance places more defects (0.85 at

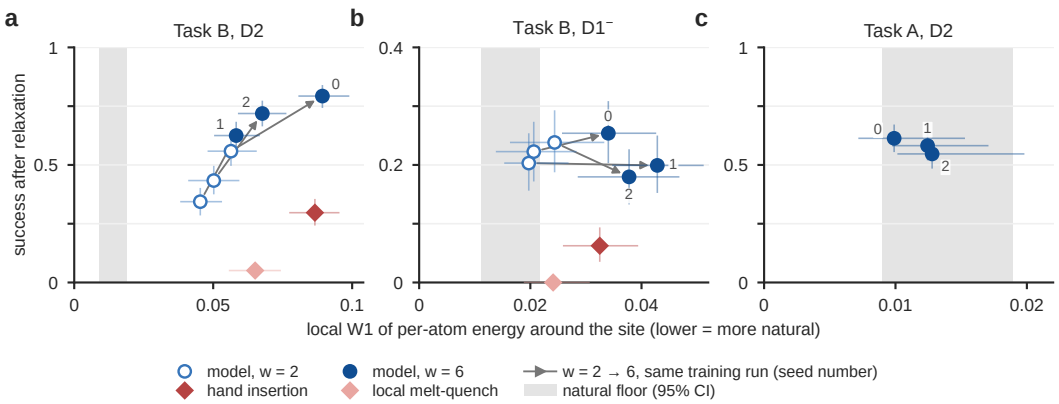


Figure 4: Success against local realism (W1) for three training runs. Open circles: $w = 2$; filled circles: $w = 6$. Each arrow joins the two results of one training run, marked with its seed number. Diamonds: baselines. Grey band: natural floor. Thin lines: 95% CIs. Panel (b) has its own success scale. For D2, every training run moves the same way as guidance gets stronger: more successes, less natural surroundings. Seed 0, used for all single-run D2 results in this paper, is the most successful run.

$w = 6$), but they sit in shallow energy minima and disappear during relaxation. Without guidance ($w = 0$), success drops (D1$^-$ 0.09 vs. 0.23, D2 0.24 vs. 0.30, at 300 steps; Appendix A).

No method can place D1$^+$, because it depends on atoms further away. Natural D1$^+$ defects survive noise and relaxation in their own glass. But when we move a natural D1$^+$ into another glass together with its 47 nearest atoms (radius 3.5σ), it disappears during relaxation 97% of the time (Appendix B). The atoms that hold it in place lie more than 3.5σ away, so neither a pasted patch nor a label on one atom can create it.

4.2 Why some generated defects survive relaxation and others do not

Natural defects are stable: after random shifts of up to about 0.19σ on average and relaxation, 99–100% of them survive (Appendix B). Generated samples move by a similar amount during relaxation (0.16–0.24σ), yet only 25–80% of their defects survive. So the problem lies with the generator: its defects sit in shallower energy minima than natural ones.

The range of training noise matters most (Table 4, Appendix A). Following DM2, we first trained with noise up to half a bond length ($\sigma_{\max} = 0.5$), but sampling starts at $\sigma = 1$, a noise level the network had never seen. Retraining with $\sigma_{\max} = 1.0$ places the D2 defect more reliably and keeps it more often after relaxation: 79% instead of 57% at $w = 6$. D2 success rises from 0.54 to 0.79 (seed 0). For D1$^-$, the first setting stays slightly better (0.22 vs. 0.21 at $w = 2$, 0.25 vs. 0.22 at $w = 6$), so we report each defect type with its better setting.

4.3 How reliable are the results?

Repeat runs and training seeds. Re-running the nine main results with the same seed gave identical numbers for eight of them, and the ninth changed within its CI. A new evaluation draw (new glasses, sites and noise) changed success by 0.02 on average. Across three training runs, D1$^-$ and Task A vary no more than this (standard deviation 0.02–0.04). Task B D2 varies about twice as much (0.08–0.11; Figure 4). The D2 runs place the defect before relaxation at different rates (0.41–0.86 at $w = 2$) and rank in the same order for success and realism, although their validation losses agree within a few percent. So each run follows the label with a different strength, and the loss does not show this. The model beats both baselines for every training run with one exception: seed 1 at $w = 2$, whose success CI overlaps that of hand insertion, although its surroundings are still clearly more natural.

Evaluation settings. We changed one setting at a time (Figure 5, Appendix A): the defect rules by ± 0.05 (with the natural references relabelled to match), the success radius by $\pm 0.1\sigma$, the realism window by $\pm 0.5\sigma$, and the rebuilt region R_{out} to 3σ and 5σ. The ranking of the methods never changed. Success moved by up to about 0.1 with the defect rules, so a rate should be reported with the rules used. Hand insertion damages only the area close to the site: its W1 falls from 0.113 to 0.068 as the window grows. The model's differences are spread over the rebuilt region (W1 0.089 at $w = 6$ for every window), so realism should be compared at more than one window size. Success changed little with R_{out} (within the CIs), although the model was trained at 4σ only, and a larger region gave much more natural D2 surroundings (W1 0.026 at $R_{\text{out}} = 5$, $w = 2$), perhaps partly because the region's edge moves out of the window.

4.4 Lessons for diffusion models of glasses

The number of sampling steps acts as a cooling rate. The sampler adds new noise at every step, so a longer run is like cooling more slowly (Table 2, Appendix A). DM2's default of 3,000 steps cools our glass too far: the energy ends below that of the training glasses, with fewer B–B contacts and a worse $g(r)$. With 100 steps the energy matches natural glass at one fifteenth of the cost, and Task B success does not change. DM2's own silica model shows the opposite problem: its samples have the right SiO_4 tetrahedra but are as disordered as glass cooled 100 times faster than its training data (Appendix C). The effective cooling rate of generated glass should be checked.

Whole generated glasses have the wrong chemistry. Even at 100 steps, generated boxes have only about two-thirds of the natural number of B–B contacts, whatever the noise range, so Task B, which keeps most of a natural glass, is the more reliable setting. In Task A the model still reaches D2 success 0.58 ± 0.03 with surroundings at the natural floor (W1 0.012 ± 0.002), against 0.34–0.35 for the Clamp and RePaint patches.

Built-in symmetry matters more than a low loss. Trained the same way at $N = 64$ (Figure 2d; Table 3, Appendix A), the EGNN makes the most natural glasses. The MPNN reaches a *lower* validation loss but makes worse glasses. The flat MLP does not learn at all: its validation loss stays at the level of always predicting zero, and 16% of its samples partly crystallise.

5 Discussion and limitations

For two of the three defect types in our model glass, a diffusion model can place a requested defect at a chosen site, with natural surroundings, more often than the methods used today. The third type, $D1^+$, cannot be placed by any method that only changes a small region, because the atoms that hold it in place lie further away.

What this means in practice. When many copies of a defect are needed in realistic surroundings, the model gives 1.5–3.5 times more correct placements per try than hand insertion, with more natural surroundings. Hand insertion stays the cheapest per try and is fine when the surroundings do not matter; re-melting a region is not a practical way to get a specific defect. Each trained model sits at its own point on the trade-off between success and realism (Figure 4), so w should be chosen for each model. Defects like $D1^+$ will need a larger rebuilt region or a label that describes the wider surroundings. Finally, checking results after relaxation, against natural floors and over several training runs and settings applies to any generative model for disordered materials; each of these checks changed a conclusion here.

Limitations and future work. Our glass is two-dimensional with only 256 atoms, and defects are defined by simple geometric rules. Whole generated glasses have the wrong chemistry, and D2 success varies between training runs (choosing w per model may help; untested). Cost matters little in 2D, where MD is cheap. We are extending the method to three-dimensional silica with a NequIP-type

network [21] and DM2's data and potential [22], aiming at defects that are rare in slowly cooled glass, such as atoms with the wrong number of bonds, non-bridging oxygens and small rings.

6 Conclusion

We built a diffusion model that can be told which atoms to keep fixed and which defect to create at a chosen site. Its network has the symmetries of the glass built in. Inside existing glasses, it places the requested defect more often than hand insertion or local melt-quench, and the new defect's surroundings are close to natural. The study also shows that the number of sampling steps, the training noise range and the training seed all change what such a model produces, so each of them should be checked rather than taken on trust.

Code and data. Our code, datasets, trained models and all results are available at `https://github.com/chilalaabubakar/IDL-PROJECT`.

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References

[1] Kai Yang and Daniel Schwalbe-Koda. A generative diffusion model for amorphous materials. *npj Computational Materials*, 12:29, 2026. doi: 10.1038/s41524-025-01901-1.

[2] Tian Xie, Xiang Fu, Octavian-Eugen Ganea, Regina Barzilay, and Tommi S. Jaakkola. Crystal diffusion variational autoencoder for periodic material generation. In *International Conference on Learning Representations*, 2022.

[3] Rui Jiao, Wenbing Huang, Peijia Lin, Jiaqi Han, Pin Chen, Yutong Lu, and Yang Liu. Crystal structure prediction by joint equivariant diffusion. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[4] Claudio Zeni, Robert Pinsler, Daniel Zügner, et al. A generative model for inorganic materials design. *Nature*, 639:624–632, 2025.

[5] Jonas A. Finkler, Yan Lin, Tao Du, Jilin Hu, and Morten M. Smedskjaer. Inverse design of amorphous materials with targeted properties. *Advanced Materials*, 38(39):e22493, 2026.

[6] Hyuna Kwon, Tim Hsu, Wenyu Sun, Wonseok Jeong, Fikret Aydin, James Chapman, Xiao Chen, Vincenzo Lordi, Matthew R. Carbone, Deyu Lu, Fei Zhou, and Tuan Anh Pham. Spectroscopy-guided discovery of three-dimensional structures of disordered materials with diffusion models. *Machine Learning: Science and Technology*, 5(4):045037, 2024.

[7] Andreas Lugmayr, Martin Danelljan, Andres Romero, Fisher Yu, Radu Timofte, and Luc Van Gool. RePaint: Inpainting using denoising diffusion probabilistic models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.

[8] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

[9] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.

[10] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

[11] Prafulla Dhariwal and Alex Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

[12] Peichen Zhong, Xinzhe Dai, Bowen Deng, Gerbrand Ceder, and Kristin A. Persson. Crystal structure prediction with host-guided inpainting generation and foundation potentials. *Materials Horizons*, 12(22):9669–9678, 2025.

[13] Jingyi Cui, Jacob K. Christopher, Ankita Biswas, Prasanna V. Balachandran, and Ferdinando Fioretto. Constrained diffusion for accelerated structure relaxation of inorganic solids with point defects. *arXiv preprint arXiv:2602.19153*, 2026.

[14] Ilia Igashov, Hannes Stärk, Clément Vignac, Arne Schneuing, Victor Garcia Satorras, Pascal Frossard, Max Welling, Michael Bronstein, and Bruno Correia. Equivariant 3D-conditional diffusion model for molecular linker design. *Nature Machine Intelligence*, 6(4):417–427, 2024.

[15] Walter Kob and Hans C. Andersen. Testing mode-coupling theory for a supercooled binary Lennard-Jones mixture I: The van Hove correlation function. *Physical Review E*, 51(5):4626–4641, 1995.

[16] Ralf Brüning, Denis A. St-Onge, Steve Patterson, and Walter Kob. Glass transitions in one-, two-, three-, and four-dimensional binary Lennard-Jones systems. *Journal of Physics: Condensed Matter*, 21(3):035117, 2009.

[17] Aidan P. Thompson et al. LAMMPS – a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Computer Physics Communications*, 271: 108171, 2022.

[18] Víctor Garcia Satorras, Emiel Hoogeboom, and Max Welling. E(n) equivariant graph neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, pages 9323–9332, 2021.

[19] Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1263–1272, 2017.

[20] Erik Bitzek, Pekka Koskinen, Franz Gähler, Michael Moseler, and Peter Gumbsch. Structural relaxation made simple. *Physical Review Letters*, 97:170201, 2006.

[21] Simon Batzner, Albert Musaelian, Lixin Sun, Mario Geiger, Jonathan P. Mailoa, Mordechai Kornbluth, Nicola Molinari, Tess E. Smidt, and Boris Kozinsky. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nature Communications*, 13: 2453, 2022.

[22] M. Bouhadja, N. Jakse, and A. Pasturel. Structural and dynamic properties of calcium aluminosilicate melts: a molecular dynamics study. *The Journal of Chemical Physics*, 138:224510, 2013.

A Additional results

Guidance weight (Task B, seed 0, 300 steps, $\sigma_{\max} = 0.5$). Success after relaxation, with the fraction placed before relaxation in brackets.

w	0	1	2	4	6
$D1^-$ success (placed)	0.09 (0.19)	0.17 (0.46)	0.23 (0.59)	0.21 (0.77)	0.23 (0.85)
D2 success (placed)	0.24 (0.32)	0.31 (0.58)	0.30 (0.73)	0.47 (0.80)	0.56 (0.97)

Task A (seed 0, 100 steps). Success after relaxation, with local W1 in brackets.

Method	D2	D1$^-$	D1$^+$
No label (chance)	0.01 (0.055)	0.00 (0.069)	0.00
Clamp patch	0.34 (0.022)	0.01 (0.053)	0.04
RePaint patch ($U = 1$)	0.35 (0.045)	0.02 (0.042)	0.04
Model $w = 6$, $\sigma_{\max} = 0.5$	0.61 (0.010)	0.23 (0.034)	0.05
Model $w = 6$, $\sigma_{\max} = 1.0$	0.84 (0.046)	0.16 (0.092)	0.01

Repeating parts of the RePaint schedule ($U = 5, 10$) does not help with this sampler. Each repeat re-runs part of the noisy schedule, so it acts as extra cooling (energy -3.704 and -3.712).

Table 2: Whole boxes generated without a label (seed 0) for different numbers of sampling steps. Natural test glasses: energy -3.681, g_{AB} floor 0.067, 0.56 B–B contacts per B atom.

noisy steps	100	300	900	2,900 (DM2)
energy per atom	-3.678	-3.692	-3.706	-3.720
$g_{AB}(r)$ L1	0.23	0.30	0.39	0.49
B–B contacts per B	0.36	0.31	0.25	—

Table 3: Symmetry test at $N = 64$ (no label, 100 steps, relaxed). Natural glasses: energy -3.636, g_{AB} floor 0.094, 0.475 B–B contacts per B atom, local W1 floor 0.007.

Network	energy	g_{AB} L1	B–B per B	local W1	crystallised	val. loss / σ^2
EGNN	-3.626	**0.21**	**0.32**	**0.044**	0%	0.23
MPNN	-3.645	0.25	0.26	0.064	0%	0.14
Flat MLP	-3.341	0.67	1.31	0.304	16%	1.02

Table 4: Training noise range (Task B, D2, seed 0). "Kept" is success after relaxation divided by the fraction placed before it.

Setting	w	placed (before relaxing)	success (after)	kept	local W1
$\sigma_{\max} = 0.5$	2	0.67	0.35	52%	0.040
$\sigma_{\max} = 0.5$	6	0.94	0.54	57%	0.064
$\sigma_{\max} = 1.0$	2	0.86	0.56	65%	0.056
$\sigma_{\max} = 1.0$	6	1.00	0.79	79%	0.089

B How stable are natural defects?

Noise, then relaxation. We shifted all atoms of 100 test glasses by random Gaussian noise, relaxed the glasses, and checked each natural defect with the success test. The table gives the fraction of defects that survived.

noise σ (mean shift)	0.05 (0.06)	0.10 (0.12)	0.15 (0.19)	0.20 (0.26)	0.30 (0.48)
D1$^-$ (155 sites)	1.00	1.00	0.99	0.68	0.08
D1$^+$ (135 sites)	1.00	0.99	0.99	0.69	0.06
D2 (1,010 sites)	1.00	0.99	1.00	0.85	0.29

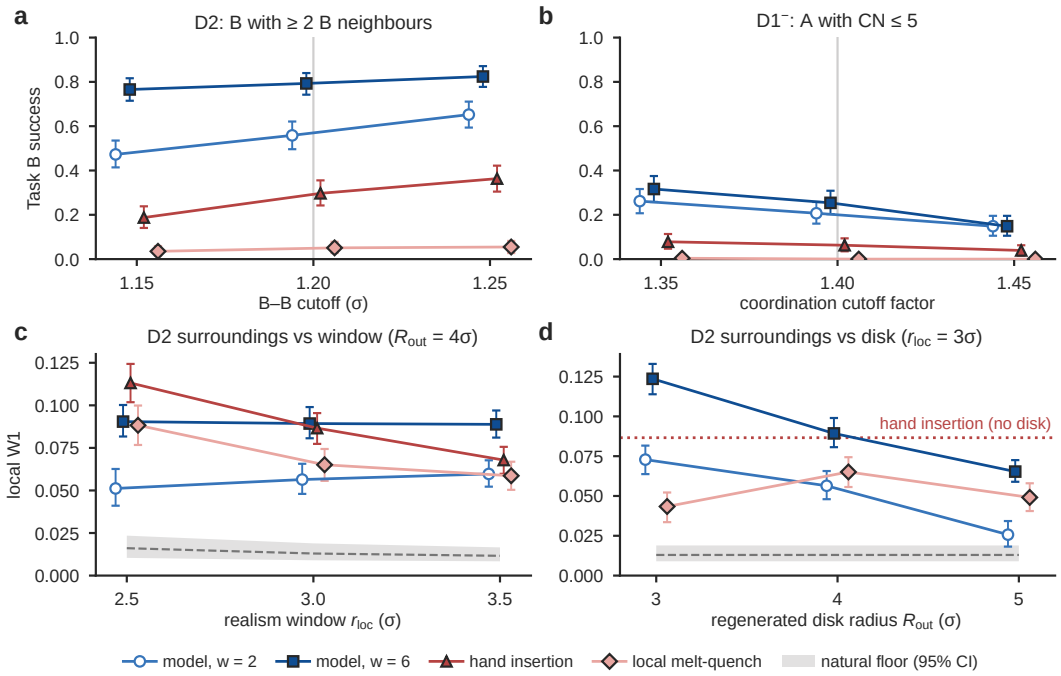


Figure 5: Changing the evaluation settings (seed 0, Task B). **(a, b)** Success for different defect rules, for D2 and D1$^-$; the natural references are relabelled with the same rules (grey line: the rule used in the paper). **(c, d)** D2 local W1 for different realism windows and rebuilt-region sizes; grey band: natural floor. The ranking of the methods never changes. Hand insertion damages only the area near the site (c), and a larger rebuilt region gives more natural surroundings (d).

Moving D1$^+$ into another glass. We pasted natural D1$^+$ defects, with their neighbours out to a given radius, into test glasses and relaxed them, 64 times for each radius. All are in place before relaxation. The fraction that survives relaxation is:

- radius 1.5σ (9 atoms): 0.00;
- radius 2.5σ (23 atoms): 0.05;
- radius 3.5σ (47 atoms): 0.03.

C Checks of earlier work

DM2 on amorphous silica. We ran DM2's trained model (no label) with its demo settings: a 300-atom box, 2,900 noisy and 100 final steps, four seeds, on CPU. We compared the samples with its training glasses, which were cooled at 1 K/ps.

- **The local structure is right.** 99.3% of Si atoms have four O neighbours, the Si–O bond is 1.650 Å (training glasses: 1.646 Å), the O–Si–O angle is 109.4°, and every peak of $g(r)$ is in place.
- **The disorder is too high.** With the same number of samples, the $g(r)$ distance to the training glasses (Si–O 0.58) is larger than between two sets of training glasses (0.31) and matches glass cooled at 100 K/ps (0.50). The spread of the O–Si–O angle (6.4° vs. 5.6°) and the number of edge-sharing tetrahedra (0.0125 vs. 0.0020 per Si) agree.

Course example. The course notebook (an MLP denoiser on a jittered 3×3 grid) reproduces the figure in our proposal over five seeds:

- pair-distance W1 0.063 (range 0.033–0.122; proposal 0.068);
- generated disorder 0.153 (proposal 0.163).

Its antisite demo trains on overlapping atoms because the site coordinates are integers. A one-line fix is described in the repository.